

1. A factorial is the product of all the integers from 1 to N . In Matlab, find the Factorial for $N = 5$.

2. Use a *for* loop to create a vector containing the first 10 elements in the harmonic series, *i.e.*,

$$\frac{1}{1} \quad \frac{1}{2} \quad \frac{1}{3} \quad \cdots \quad \frac{1}{10}$$

3. Use a *for* loop to create a vector containing the first 10 elements in the alternating harmonic series, *i.e.*,

$$\frac{1}{1} \quad -\frac{1}{2} \quad \frac{1}{3} \quad -\frac{1}{4} \cdots -\frac{1}{10}$$

4. Write Matlab code to calculate the sum of series:

$$S = 1 - \frac{x}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} \quad \text{for } x = 1.5.$$

5. Write the Matlab program to obtain the sum of all even number from 0 to 20 using *for* loop statement.

6. Write the program to find the average value of given any 10 number using *for* loop statement. Check the program with the following values : 35, 24, 5, 6, 4, 10, 23, 45, 2

7. Repeat the preceding problem, this time using a *while* loop.

8. Write a program, using *while* loop, for finding square of integers less than 5.

9. Write the Matlab program, to test whether or not π^e is greater than, or equal to, e^π .

10. In Matlab, compute the Fibonacci sequence defined by $f_1 = 0, f_2 = 1$ and $f_n = f_{n-1} + f_{n-2}$. Using *for* loop and *if* condition.

11. Draw graphs of $\sin(nx)$ on the interval $[-1, 1]$ for $n = 1, 2, 3, 4, 5, 6, 7, 8$.

12. For any n build the $n \times n$ matrix

$$A = \begin{bmatrix} 1 & 0 & 0 \cdots & 0 \\ 1 & 1 & 0 \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & 1 & 1 \end{bmatrix}$$

in Matlab.